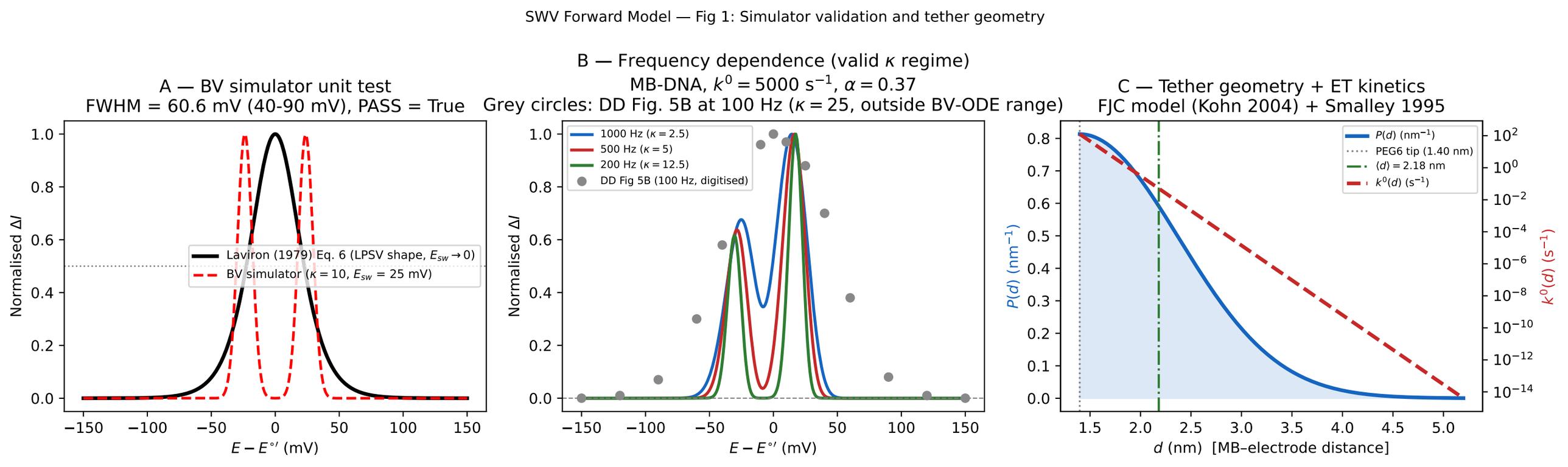
SWV Forward Model — Fig 1: Simulator validation and tether geometry
A — BV simulator unit test
FWHM = 60.6 mV (40-90 mV), PASS = True
Laviron (1979) Eq. 6 (LPSV shape, $E_{sw} \to 0$)
BV simulator ($\kappa = 10$, E_{sw} = 25 mV)
Normalised ΔI
$E - E^{\circ\prime}$ (mV)
B — Frequency dependence (valid κ regime)
MB-DNA, $k^0 = 5000$ s^{-1}, $\alpha = 0.37$
Grey circles: DD Fig. 5B at 100 Hz ($\kappa = 25$, outside BV-ODE range)
1000 Hz ($\kappa = 2.5$)
500 Hz ($\kappa = 5$)
200 Hz ($\kappa = 12.5$)
DD Fig 5B (100 Hz, digitised)
Normalised ΔI
$E - E^{\circ\prime}$ (mV)
C — Tether geometry + ET kinetics
FJC model (Kohn 2004) + Smalley 1995
$P(d)$ (nm^{-1})
PEG6 tip (1.40 nm)
$\langle d \rangle = 2.18$ nm
$k^0(d)$ (s^{-1})
$P(d)$ (nm^{-1})
$k^0(d)$ (s^{-1})
d (nm) [MB–electrode distance]